Math 344, S19 Name: Answer

Quiz 4

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given

$$\mathbf{A} = \begin{bmatrix} 2 & 4 & 6 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}.$$

Find $\mathbf{A}^{-1}$ using the Gaussian-Jordan Elimination.

$$[A \,|\, I] = \left[\begin{array}{ccc|ccc} \boxed{2} & 4 & 6 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R2 \leftarrow R2 - R1} \left[\begin{array}{ccc|ccc} \boxed{2} & 4 & 6 & 1 & 0 & 0 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R3 \leftarrow R3 - (\frac{1}{2})R1} \left[\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 0 & \boxed{1} & -3 & -1 & 1 & 0 \\ 0 & -2 & 5 & -\frac{1}{2} & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R3 \leftarrow R3 + 2R2} \left[\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 0 & 0 & \boxed{-1} & -\frac{5}{2} & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R2 \leftarrow R2 - 3R3} \left[\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{13}{2} & -5 & -3 \\ 0 & 0 & \boxed{-1} & -\frac{5}{2} & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R1 \leftarrow R1 + 6R3} \left[\begin{array}{ccc|ccc} 2 & 4 & 0 & -14 & 12 & 6 \\ 0 & \boxed{1} & 0 & \frac{13}{2} & -5 & -3 \\ 0 & 0 & -1 & -\frac{5}{2} & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R1 \leftarrow R1 - 4R2} \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & -40 & 32 & 18 \\ 0 & 1 & 0 & \frac{13}{2} & -5 & -3 \\ 0 & 0 & -1 & -\frac{5}{2} & 2 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -20 & 16 & 9 \\ 0 & 1 & 0 & \frac{13}{2} & -5 & -3 \\ 0 & 0 & 1 & \frac{5}{2} & -2 & -1 \end{array}\right]$$

$$= [I \,|\, A^{-1}]$$

Thus $A^{-1} = \begin{bmatrix} -20 & 16 & 9 \\ \frac{13}{2} & -5 & -3 \\ \frac{5}{2} & -2 & -1 \end{bmatrix}$

Problem 2. (5 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 3 & -5 \\ 1 & 2 & -7 \\ 3 & 9 & -4 \end{bmatrix}$. Find the **LU** and **LDU** factorizations of **A**.

$$A = \begin{bmatrix} \boxed{2} & 3 & -5 \\ 1 & 2 & -7 \\ 3 & 9 & -4 \end{bmatrix} \xrightarrow{\ell_{21}=\frac{1}{2}} \begin{bmatrix} \boxed{2} & 3 & -5 \\ 0 & \frac{1}{2} & -\frac{9}{2} \\ 3 & 9 & 4 \end{bmatrix}$$

$$\xrightarrow{\ell_{31}=\frac{3}{2}} \begin{bmatrix} 2 & 3 & -5 \\ 0 & \boxed{\frac{1}{2}} & -\frac{9}{2} \\ 0 & \frac{9}{2} & \frac{7}{2} \end{bmatrix}$$

$$\xrightarrow{\ell_{32}=9} \begin{bmatrix} 2 & 3 & -5 \\ 0 & \frac{1}{2} & -\frac{9}{2} \\ 0 & 0 & 44 \end{bmatrix} = U$$

Thus

$$A = \overset{L}{\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{3}{2} & 9 & 1 \end{bmatrix}} \overset{U}{\begin{bmatrix} 2 & 3 & -5 \\ 0 & \frac{1}{2} & -\frac{9}{2} \\ 0 & 0 & 44 \end{bmatrix}}$$

$$= \overset{L}{\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ \frac{3}{2} & 9 & 1 \end{bmatrix}} \overset{D}{\begin{bmatrix} 2 & & \\ & \frac{1}{2} & \\ & & 44 \end{bmatrix}} \overset{U}{\begin{bmatrix} 1 & \frac{3}{2} & -\frac{5}{2} \\ 0 & 1 & -9 \\ 0 & 0 & 1 \end{bmatrix}}$$